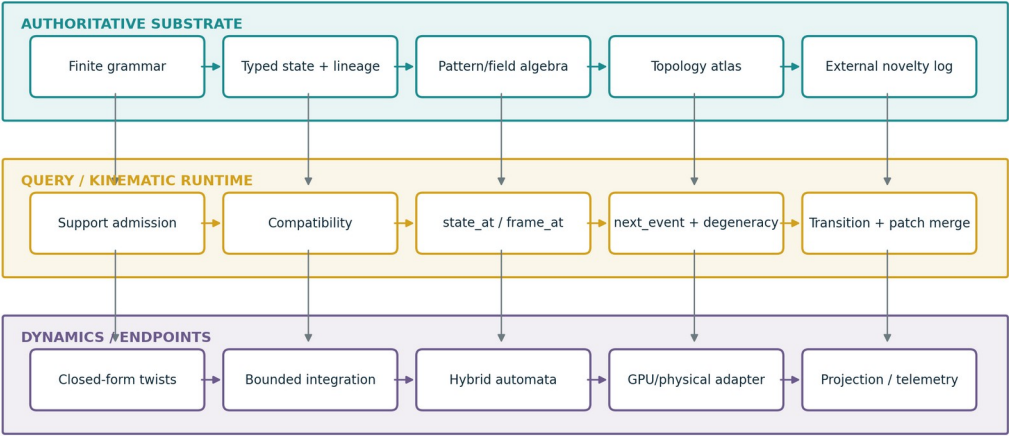


UNIFIED GEOMETRIC- TOPOLOGICAL SUBSTRATE

Pattern, Kinematic Calculus and Dynamics Expansion
UGTS-KC 2.0

UGTS-KC 2.0 expanded query-first architecture



Authority remains support -> compatibility -> guard/event -> transition -> lineage. Patterns and dynamics extend the operator set; they do not replace evidence boundaries.

Figure 1. Expanded architecture: pattern and dynamic operators extend the canonical query-first substrate.

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Requester-supplied identifier	NL200678942
Requester-supplied date of birth	10 July 1990
Version	2.0.0
Generated	16 August 2026
Status	Engineering expansion with executable reference validation

ATTRIBUTION AND LEGAL BOUNDARY

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Contents and reading rule

This upgrade is designed to be read as an extension of the supplied UGTS-0 and UGTS-GN 1.1 reports. It preserves their evidence boundary and query-first event architecture while adding a larger, executable pattern and kinematic calculus layer.

- 1. Executive upgrade decision
- 2. Extended substrate architecture
- 3. Parametric pattern calculus
- 4. Field and surface pattern calculus
- 5. Topology expansion
- 6. Kinematic calculus
- 7. Dynamics and hybrid event resolution
- 8. Integration, GPU mapping and memory
- 9. Reference implementation and validation
- 10. Boundaries and kill criteria
- 11. Final upgraded definition
- Appendix A. M198-M257 extended mechanism catalog
- Appendix B. API and exchange dictionary
- Appendix C. Source and package verification

READING RULE

The core is still not a renderer. Geometry supplies typed patterns, fields and local supports; topology supplies explicit gluing and route classes; kinematics supplies state_{at(t)}, frames and feasible motion; dynamics supplies bounded flows; events remain compatibility-gated guard crossings with lineage.

Document control

Source basis	Pages	Filename	SHA-256 prefix	Treatment
UGTS-0 1.0	45	Unified_Geometric_Topological_Substrate(1).pdf	c991aba24cc939ba...	Referenced by hash; not redistributed
UGTS-GN 1.1	28	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum(2).pdf	3a6b6c57118ca4e2...	Referenced by hash; not redistributed

1. Executive upgrade decision

1.1 Baseline retained

The supplied reports converge on one stable architecture: a finite grammar generates typed state; local radial-angular support removes irrelevant candidates; compatibility determines whether states may couple; an explicit relation or guard defines an event surface; the earliest valid event applies route, topology and lineage updates; irreducible novelty is logged. UGTS-KC 2.0 keeps that pipeline unchanged.

The GPU-native addendum freezes a 197-mechanism normalized baseline. This extension does not rewrite those entries. It appends 60 engineering mechanisms, M198-M257, and labels them as additions rather than pretending they were all source-derived.

Item	Count/status	Meaning
Baseline catalog	197 mechanisms	Existing source-derived and normalized operators
Engineering additions	60 mechanisms	20 pattern, 12 field/surface, 10 topology, 12 kinematic, 6 dynamics/event
Extended total	257 mechanisms	Baseline plus this versioned extension
Reference tests	47 passing	Standard-library Python implementation
Physical claims	None promoted	Hardware advantage still requires target measurement

1.2 What is new

- A versioned parametric pattern algebra for superellipses, superformula families, harmonic curves, trochoids, spirals, splines and constant-width profiles.
- A field/surface layer for superquadrics, swept tubes, ruled surfaces, helicoids, catenoids, gyroid/Schwarz periodic fields, compact blends, offsets and arrival-time fields.
- Topology descriptors for winding, covering lifts, homotopy signatures, braid words, cell incidence, discrete Morse critical cells and atlas cocycles.
- SE(2)/SE(3) twist calculus, quaternion and screw interpolation, Frenet/Bishop frames, arc-length maps, curvature limits, jerk-bounded schedules and articulated kinematics.
- Bounded dynamics for symplectic integration, damped modes, graph diffusion, reaction-diffusion, hybrid automata and explicit event degeneracy resolution.

1.3 Upgrade decision

DECISION

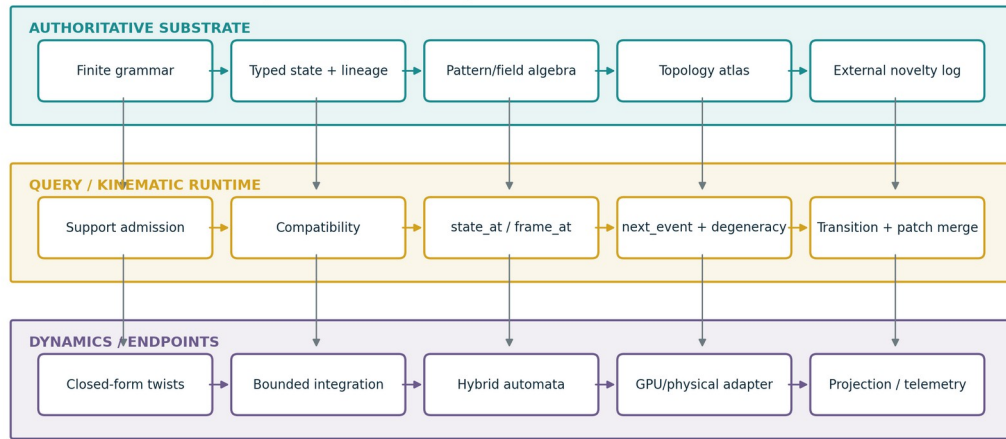
GO for a pattern-and-kinematics expansion, provided every new shape is a typed primitive or field, every motion has a parameter and error contract, every topology label has an explicit map or invariant, and every dynamic mode still commits through support, compatibility, guard and lineage.

1.4 Concrete deliverables

- This upgraded PDF and editable DOCX source.
- A machine-readable 60-mechanism catalog in CSV and JSON.
- A JSON Schema and example expanded substrate definition.
- A dependency-free Python reference package covering patterns, fields, topology, kinematics, dynamics and events.
- 47 passing executable tests with captured results.
- Original diagrams, source hashes, file hashes and a reproducible ZIP manifest.

2. Extended substrate architecture

UGTS-KC 2.0 expanded query-first architecture



Authority remains support -> compatibility -> guard/event -> transition -> lineage. Patterns and dynamics extend the operator set; they do not replace evidence boundaries.

Figure 2. The authoritative path remains query-first. Pattern, kinematic and dynamic operators are typed extensions, not a replacement for the core event calculus.

2.1 Extended state

The compact baseline state remains valid. Pattern-aware motion adds only the fields needed by a selected profile; unused dynamics do not have to occupy the hot record.

```

q = (position, time, phase, sheet, orientation,
     generative_address, branch, policy, uncertainty,
     linear_velocity, angular_velocity,
     curvature, torsion, mode, dynamic_state)
  
```

Coordinates remain non-authoritative for identity. A repeated point on a rose curve, torus-like route or quotient chart can represent different times, sheets, windings, branches or lineage addresses.

2.2 Pattern instance contract

```

pattern = (mechanism_id, parameters, frame,
           support_id, compatibility_tags,
           validation_contract)

validation_contract = (domain, units, parameter bounds,
                      derivative availability,
                      solver family, error budget)
  
```

A pattern is therefore not merely a name such as "spiral". It is a versioned operator with declared scale, phase range, frame, derivatives and validation limits. The same curve may be used as geometry, a local support, a trajectory, a guard, a projection path or a calibration sweep, but the role is explicit.

2.3 Query set

Query	Contract
state_at(entity,t)	Evaluate closed-form or bounded integrated state.
frame_at(pattern,s or t)	Return Frenet/Bishop/body frame and status.
curvature_at(pattern,s)	Return curvature/torsion or a bounded numerical estimate.
next_event(entity,t0,t1)	Solve the earliest supported compatible guard event.
events_in_support(support,t0,t1)	Return events whose pattern or field support intersects the query.

Query	Contract
route_class(path)	Return winding/homotopy/braid/cell descriptor.
reconstruct_identity(entity)	Rebuild lineage from seed, transitions and novelty log.

2.4 Verified-event rule

```
verified = in_support
    and compatible
    and guard_status in accepted_statuses
    and confidence >= confidence_floor
    and numeric_error <= event_margin

accepted_statuses may include crossing, touch or tangency,
but the policy must be declared before execution.
```

NON-NEGOTIABLE BOUNDARY

A curve self-intersection, a zero of a periodic field or a topological crossing is not automatically an interaction. Sheet, phase, route class, mode, policy and lineage still determine compatibility.

3. Parametric pattern calculus

Expanded parametric pattern families

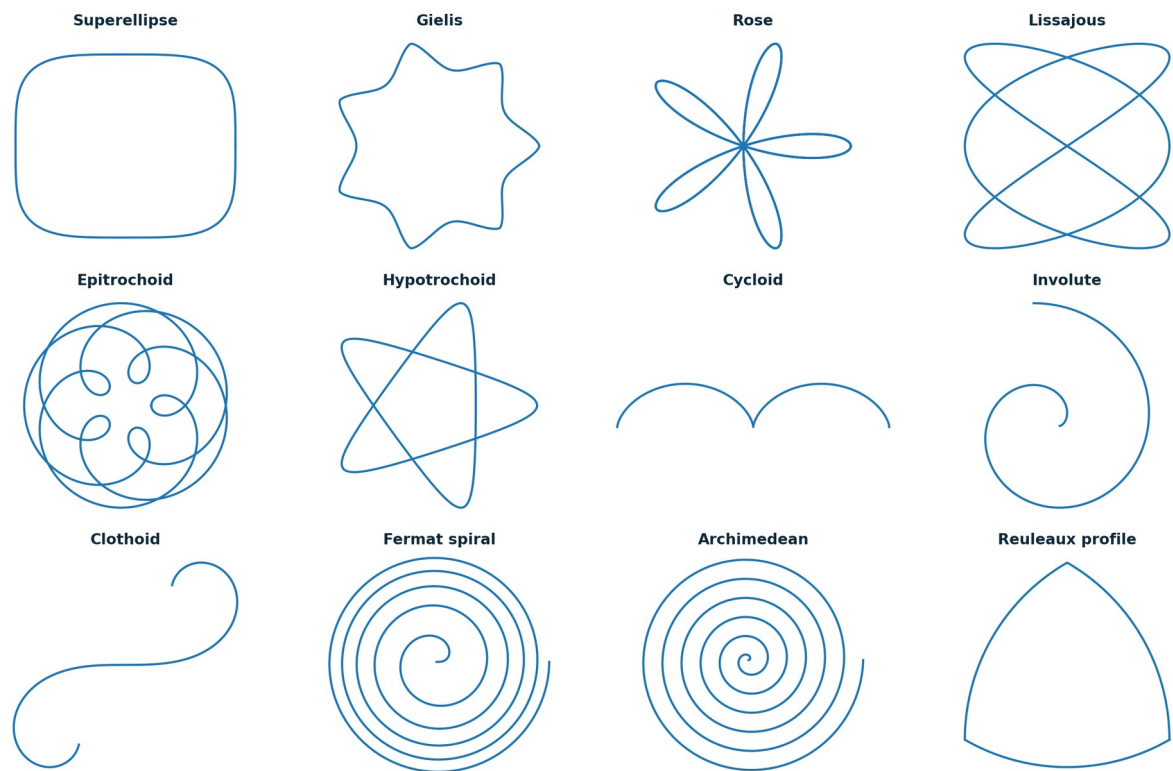


Figure 3. Representative M198-M217 pattern families. The gallery is generated from the bundled reference functions.

3.1 Family taxonomy

Family	Mechanisms	Substrate role
Exponent and radial families	Superellipse, Gielis superformula, rose	Compact template grammar; parameter singularities are bounded.
Harmonic and rolling families	Lissajous, epitrochoid, hypotrochoid, cycloid, trochoid	Useful for periodic route, cusp, contact and calibration tests.

Family	Mechanisms	Substrate role
Unwrapping and equilibrium	Involute, catenary, clothoid	Nonuniform curvature and mechanically interpretable shapes.
Spiral families	Fermat, Archimedean	Address/shell sweeps with explicit growth law.
Spline families	Rational quadratic, cubic Bezier, B-spline, NURBS	Local control, bounded degree and explicit knot/weight schemas.
Partition and skeleton families	Reuleaux profile, power cells, sampled medial axis	Constant clearance, routing cells and compressed shape descriptors.

3.2 Pattern grammar

The pattern grammar extends the finite grammar already present in UGTS. Each production emits a normalized primitive and optional derivatives rather than unrestricted symbolic text.

```

PatternExpr := Primitive(parameters)
              | Transform(frame, PatternExpr)
              | Restrict(domain, PatternExpr)
              | Repeat(period, count, PatternExpr)
              | Sweep(profile, centerline, frame_policy)
              | FieldGuard(field, threshold, band)

Hard limits: max_depth, max_symbols, max_control_points,
             max_period_cells, max_derivative_order.

```

3.3 Derivative and event semantics

For a parametric curve $r(t)$, the first and second derivatives determine speed, tangent and curvature. Cusps occur when $|r'(t)|$ approaches zero; self-intersections require both coordinate equality and distinct parameter or lineage labels. The event solver must not silently divide by zero or merge branches.

```

speed(t)      = |r'(t)|
curvature(t)  = |r'(t) x r''(t)| / |r'(t)|^3    (3D)
cusp status   = speed(t) <= speed_floor
crossing      = r(t1) ~= r(t2), t1 != t2, plus compatibility policy

```

3.4 Selected technical notes

- The Gielis superformula is admitted as a bounded radial grammar, not as a universal biological law.
- The clothoid is numerically integrated in the bundled code; its step count and position error belong to the validation contract.
- A generic Bezier or spline blend is not automatically arc-length parameterized and not automatically an exact signed-distance boundary.
- A Reuleaux profile has constant width under its ideal construction; discretized sampling introduces chord error that must be budgeted.
- Voronoi/power cells are routing partitions. They do not replace compatibility, uncertainty or lineage.

4. Field and surface pattern calculus

Implicit field and surface-pattern slices

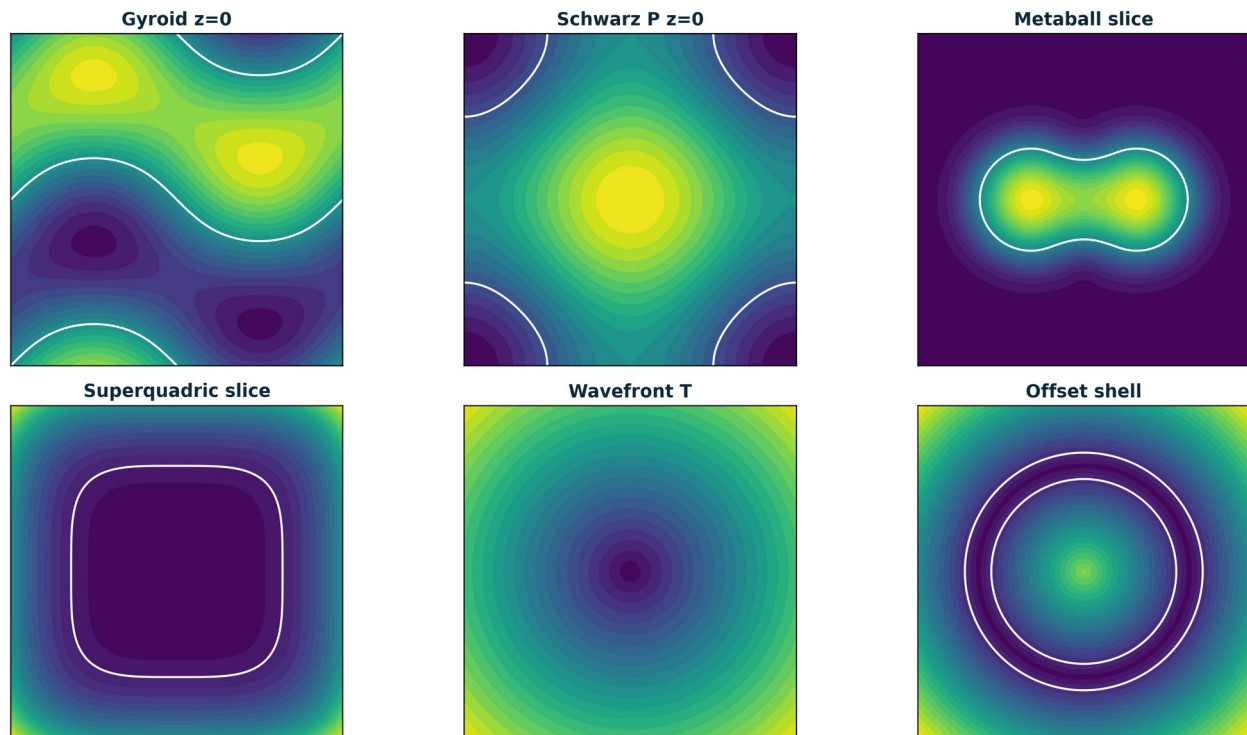


Figure 4. Representative implicit fields and zero contours. Zero is an event/transition candidate only under the declared guard policy.

4.1 Field rule

The baseline report distinguishes exact signed-distance functions from generic implicit fields. UGTS-KC 2.0 keeps that distinction. Superquadrics, periodic trigonometric fields and compact blends are generally implicit fields; they may be used for sign, guard and gradient queries without being mislabeled as exact Euclidean distance.

Field family	Use	Boundary
Superquadric / ellipsoid	Shape and directional metric support	Check positive scales, exponent range and conditioning.
Swept tube / cylinder	Distance to centerline or transported profile	Sample error and moving-frame choice are explicit.
Ruled surface / helicoid / catenoid	Sheet, portal or neck geometry	Mathematical surface properties do not imply physical self-assembly.
Gyroid / Schwarz-P	Periodic field template	Threshold, finite cell count and boundary conditions are required.
Metaball blend	Smooth merge/split benchmark	Kernel support and saturation prevent long-range hidden coupling.
Offset / arrival field	Tolerance band and event-time support	Exact offset requires distance-like input; variable speed requires a solver.

4.2 Sweeps and moving frames

A generalized cylinder transports a profile along a centerline. The frame policy is part of the schema because Frenet frames can become unstable near zero curvature, while Bishop frames trade torsion tracking for rotation-minimizing transport.

$$X(s, u, v) = c(s) + u e_1(s) + v e_2(s)$$

```
frame_policy = "frenet"  when curvature > kappa_floor
                | "bishop" for rotation-minimizing transport
                | "body"   for externally supplied orientation
```

4.3 Periodic-field topology

The topology of a gyroid-like or Schwarz-P field depends on threshold, domain and boundary identification. The package therefore stores the field formula, threshold, cell bounds and chart/gluing rules separately. A visual resemblance to a labyrinth does not establish a material or energy property.

PHYSICAL BOUNDARY

Minimal-surface names describe mathematical templates. A manufactured structure still requires material constitutive laws, loads, tolerances, losses and measured stability.

5. Topology expansion

Topology descriptors added without replacing geometry or lineage

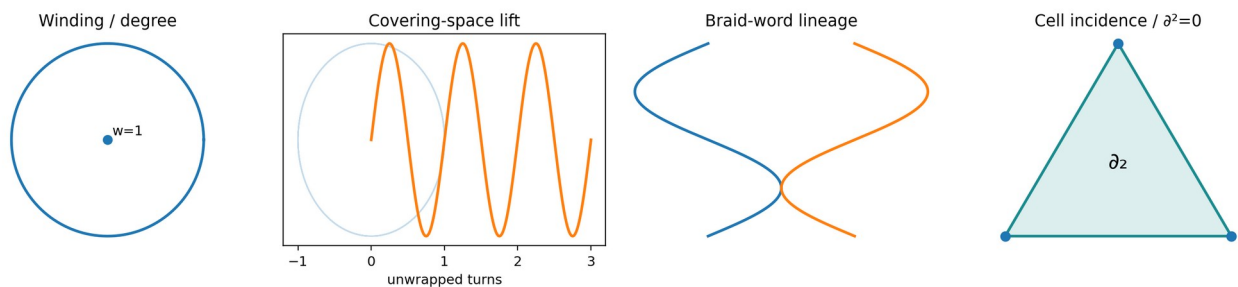


Figure 5. Winding, covering, braid and cell-complex descriptors enrich route identity while keeping geometry and lineage separate.

5.1 Winding, covering and homotopy

A periodic or self-crossing route cannot be identified by coordinates alone. The covering-space lift records traversal count, and the homotopy signature records winding around declared obstacles. These descriptors become compatibility tags and lineage fields, not replacements for the path.

```
theta_lift = theta_mod + 2*pi*winding
route_class = (winding_obstacle_1, ..., winding_obstacle_n)

same_position and different route_class -> distinct state sectors
```

5.2 Braid words and multi-entity lineage

Ordered crossings of multiple routes can be recorded by braid generators. The reference reducer only cancels adjacent inverse pairs; it does not claim a complete canonical braid normal form. Word length and generator count are bounded to prevent lineage explosion.

5.3 Cell complexes and incidence invariants

A cell complex stores topology independent of embedding. Oriented boundary maps must satisfy boundary-of-boundary zero. Any portal, split or merge edit that violates this invariant is rejected or logged as an unresolved topology event.


```
boundary_k : C_k -> C_(k-1)
  boundary_(k-1) * boundary_k = 0
```

```
topology edit = incidence patch + lineage patch + validation result
```

5.4 Discrete Morse and persistence descriptors

Discrete Morse critical cells and persistent-homology barcodes are admitted only as bounded diagnostic descriptors. The ZIP defines their exchange form, but it does not bundle a certified persistent-homology solver. They cannot replace exact transition records or full geometry.

5.5 Atlas transition cocycle

Multi-chart portals must agree on overlaps. The cocycle check prevents a route from receiving inconsistent coordinates or orientation when composed through different chart sequences.

```
g_ij * g_jk = g_ik on every declared triple overlap
```

6. Kinematic calculus

Frenet/Bishop frame transport and SE(3) sweep orientation

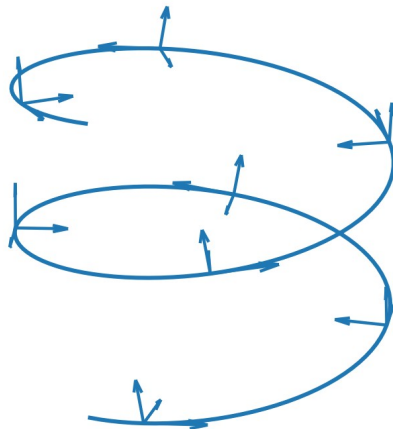


Figure 6. Moving frames transport profile orientation along a 3D curve. Frame status is part of the query result.

6.1 SE(2) and SE(3) state-at-time

Rigid motion is represented on SE(2) or SE(3), not by adding unrelated position and angle values. Constant body twists have closed-form exponentials, supporting horizon-skipping state_at(t) for a restricted motion family.

```
planar: T(t) = exp(t * twist(vx, vy, omega))
spatial: T(t) = exp(t * twist(omega_vector, v_vector))
```

```
state_at(t) cost follows fixed expression size,
not the number of omitted display frames.
```

6.2 Orientation interpolation

Quaternion SLERP handles unit orientation with shortest-arc and antipodal policies. Screw interpolation uses the relative SE(3) logarithm/exponential so translation and rotation evolve as one rigid transform. Neither operation changes sheet, branch or lineage unless an explicit transition says so.

6.3 Frenet and Bishop frames

Frame	Strength	Boundary
Frenet-Serret	Uses first/second derivatives; exposes curvature and binormal.	Fails or becomes unstable when speed/curvature is near zero.
Bishop / parallel transport	Rotation-minimizing frame across sampled points.	Requires initial normal and numerical transport tolerance.
Body/external frame	Orientation supplied by state or actuator.	Must be normalized and compatible with the sweep frame.

6.4 Arc length and feasible steering

Arc-length tables provide monotone lookup for uniform-speed travel. Curvature limits convert path shape into a speed and angular-rate constraint before event solving.

```
omega = v * kappa
v <= sqrt(a_lateral_max / max(|kappa|, epsilon))
```

6.5 Jerk-bounded timing

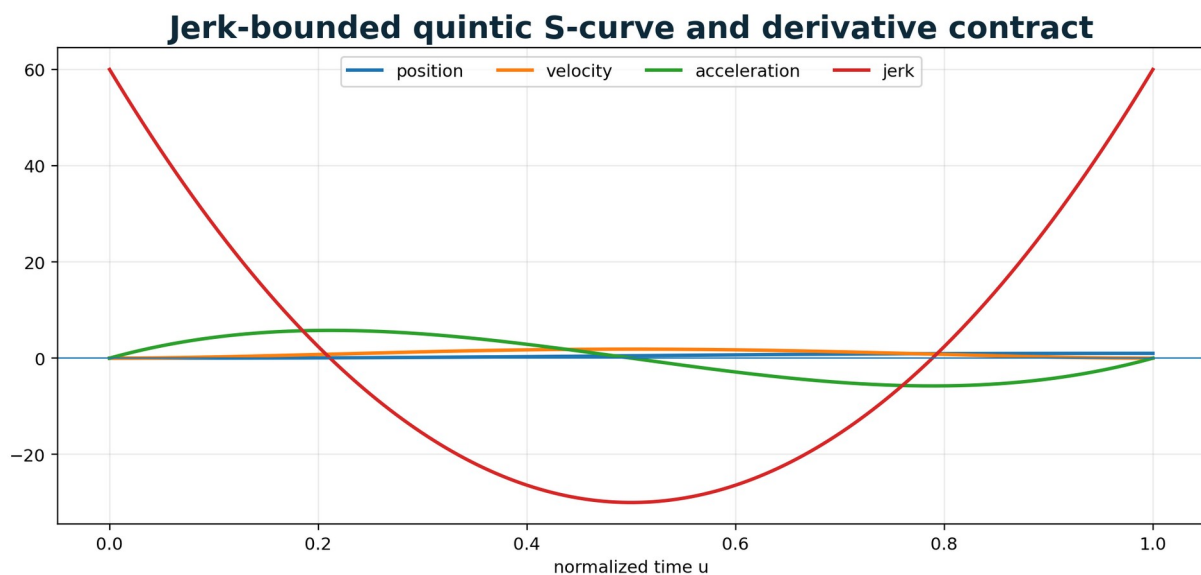


Figure 7. Quintic S-curve. Position, velocity and acceleration are continuous; peak derivatives define the minimum duration contract.

The bundled time law uses $s(u)=10u^3-15u^4+6u^5$. It is smooth and jerk-bounded over a finite duration, but it is not advertised as globally time-optimal for every path. The duration is enlarged until velocity, acceleration and jerk limits all hold.

6.6 Articulated chains

Forward kinematics composes joint transforms. The optional inverse solver uses damped least squares and reports residual and iteration budget. Singular or unreachable targets return a bounded failure state rather than an invented solution.

7. Dynamics and hybrid event resolution

7.1 Bounded dynamic operators

Operator	Role	Boundary
Symplectic Euler	Fallback state integration with explicit dt.	Qualitative energy behavior improves, but numerical error remains.
Damped second-order mode	Mass-spring-damper actuator/hinge/reset.	Mass, stiffness, damping, forcing and energy boundary are typed.
Graph diffusion	Local exchange over explicit adjacency.	Stability depends on alpha, dt and graph degree.
Reaction-diffusion	Gray-Scott two-field pattern update.	Grid, boundary condition, coefficients and time step are explicit.
Hybrid automaton	Continuous mode flows plus guard/reset transitions.	Mode, guard, reset, priority and lineage are versioned.

7.2 Hybrid calculus

The source threshold/cup and parity-hinge patterns become special cases of a hybrid automaton. A mode carries continuous dynamics; a guard triggers an event; a reset updates state, route and lineage. This unifies static geometry events with dynamic thresholds without confusing the field with the transition.

```
x' = f_mode(x)
if support and compatibility and g_mode, target(x) crosses 0:
    x_plus = reset_mode, target(x_minus)
    append(event, route, lineage, confidence)
```

Hybrid event calculus with explicit tangency and equal-time handling



crossing | touch | tangency | coincide | priority -> commutative merge -> conflict event

Figure 8. Equal-time and tangency handling are explicit stages before an authoritative patch is committed.

7.3 Event degeneracy classes

Status	Meaning	Resolution
crossing	Guard changes sign.	Commit when support, compatibility and confidence hold.
touch	Guard reaches band but derivative indicates no certified crossing.	Policy may log or reject.
tangency	Guard is near zero and first derivative is near zero.	Use higher-order or interval status; avoid false toggle.
coincident	Guard stays within tolerance over an interval.	Require separate constraint mode; do not emit infinite events.

Status	Meaning	Resolution
simultaneous	Two certified event-time intervals overlap.	Apply priority, commutative merge, or conflict event.

CHATTER AND ZENO BOUNDARY

Hysteresis, minimum hold time and event coalescing are mandatory where repeated guard crossings could accumulate. The substrate must report event density and stop when ordering or storage becomes unbounded.

8. Integration, GPU mapping and memory

8.1 Canonical integration path

```

pattern parameters
  -> local support pruning
  -> topology / mode / policy compatibility
  -> kinematic state and frame query
  -> guard value + derivative + error interval
  -> degeneracy resolver
  -> verified event
  -> route / topology / dynamic reset
  -> lineage + novelty log

```

8.2 GPU-native mapping

The existing G64/E32 and G32/E16 profiles can be extended with pattern opcode and parameter-buffer references rather than inflating every hot record. A production GPU path would group candidates by opcode, evaluate derivatives in coherent kernels and compact only verified events.

GPU field/path	Purpose	Boundary
Pattern opcode	u16/u32 enum selecting a versioned primitive.	Unknown opcodes reject with reason code.
Parameter reference	Offset into aligned parameter buffer.	Bounds and schema hash checked before dispatch.
Derivative flags	Availability of r , r' , r'' , gradient or Hessian.	Missing derivatives force a declared numerical path.
Frame/status output	Tangent/frame, curvature, residual, degeneracy.	Packed only if error remains below event margin.
Compact event append	Verified events only.	Contention and compaction cost must be benchmarked.

PERFORMANCE BOUNDARY

This extension does not claim new Vulkan, FPGA, ASIC or photonic throughput. The Python tests establish functional consistency only. Any GPU or physical performance statement requires target-specific compilation, error comparison and measurement.

8.3 Memory and reconstructibility

Closed pattern parameters and analytic twists may be reconstructed from seed and grammar; integrated dynamics, external inputs, mode changes, solver ambiguities and calibration changes require authoritative event or checkpoint records. Novelty-only retention remains valid only when reconstruction is demonstrated.

- Store pattern parameters once and reference them from instances.

- Store sampled arc-length tables or sweep frames only when their recomputation cost or determinism requires it.
- Log every topology edit, mode reset, external force/calibration change and conflict resolution.
- Treat compact winding, braid or lineage hashes as checksums, not complete identity.
- Measure event/branch density before claiming memory advantage.

8.4 Physical-device boundary

A physical endpoint may implement selected patterns as trajectories, masks, waveguide layouts or actuation schedules. Geometry/topology labels remain routing abstractions unless the relevant Maxwell, fluid, solid, thermal and control equations are solved or calibrated. Full-system energy, drift, latency and false/missed-event rates remain required metrics.

9. Reference implementation and validation

9.1 Package layout

Path	Purpose
report/	PDF and editable DOCX source.
spec/	Catalog CSV/JSON, schema, state/event notes and source hashes.
src/ugts_kc/	Dependency-free reference implementation.
tests/	47 unit tests covering geometry, topology, kinematics, dynamics and events.
examples/	Expanded JSON definition and runnable demos.
diagrams/	Original report figures generated from the code.
validation/	Captured test results, counts and SHA-256 manifest.
docs/	Changelog, evidence boundary and attribution notice.

9.2 Implemented modules

Module	Responsibility
patterns.py	M198-M217 parametric curves, splines, partition and skeleton helpers.
fields.py	M218-M229 implicit fields, sweeps and arrival functions.
topology.py	Winding, covering, homotopy, braid, cell complex and cocycle checks.
kinematics.py	SE(2)/SE(3), SLERP, screw interpolation, frames, timing and chain kinematics.
dynamics.py	Symplectic, damped, graph, reaction-diffusion and hybrid automaton operators.
events.py	Root bracketing, degeneracy classification and equal-time patch resolution.

9.3 Test result

VALIDATION RESULT

47 tests ran under Python 3.13.5; status: PASS. The captured console output is stored in validation/test_results.txt.

The tests cover endpoint identities, periodicity, field signs, spline/NURBS behavior, winding and boundary invariants, SE(2)/SE(3) transforms, moving frames, S-curve limits, forward/inverse kinematics, integration, diffusion, reaction-diffusion, hybrid transitions, root solving and patch conflicts. They establish internal consistency of the bundled implementation, not external performance or complete numerical certification.

9.4 Reproduction

```
python -m unittest discover -s tests -v
PYTHONPATH=src python examples/kinematic_event_demo.py

# Optional: inspect the machine-readable definition
python -m json.tool examples/expanded_substrate_definition.json
```

9.5 Validation interpretation

Validation layer	What is established	Status
Functional	Bundled functions satisfy the included unit properties.	PASS
Catalog traceability	M198-M257 count and IDs are machine checked.	PASS
Schema	Example JSON validates structurally against the supplied contract by inspection; no external validator is required by the runtime.	SUPPLIED
Numerical certification	Most functions use ordinary floating point and bounded sampling.	APPLICATION-SPECIFIC
GPU/physical performance	No target benchmark was added in this extension.	NOT CLAIMED
Legal authorship/identity	Requester attribution is recorded but not independently verified.	NOT ESTABLISHED

10. Boundaries and kill criteria

10.1 Numerical kill criteria

- Pattern parameter singularities, spline denominators or frame normalization fall below declared numerical floors.
- Sampled clothoid, tube, arc-length or medial-axis error exceeds the guard/event margin.
- Curvature or torsion becomes ill-conditioned and no fallback frame/status is accepted.
- Equal-time event intervals cannot be ordered or merged without conflicting authoritative patches.
- Time step, graph degree or reaction-diffusion coefficients violate the selected stability budget.

10.2 Topology and lineage kill criteria

- A topology edit violates boundary-of-boundary zero, chart cocycle consistency or declared orientation maps.
- Winding, braid, homotopy or persistent descriptors grow faster than the avoided materialization.
- Split/merge ancestry cannot be reconstructed or compact checksums collide within the application budget.
- Coordinates are used as identity after self-intersection, quotient wrapping or route re-entry.

10.3 Performance kill criteria

- Pattern opcode divergence, derivative evaluation, root solving or event compaction dominates throughput.
- Compatibility pruning is too weak to offset the cost of the richer pattern set.

- Integrated dynamics or dense reaction-diffusion state removes the expected memory advantage.
- A conventional geometry/physics/indexing implementation is cheaper, faster or more accurate at equal error.

10.4 Physical and claim boundary

CLAIMS NOT PROMOTED

No zero memory, zero heat, zero latency, universal $O(1)$, topology-powered energy, physical Klein self-assembly, exact arbitrary-field solver or one-bit-complete-state claim is introduced. Geometry/topology vocabulary remains subordinate to typed state, measured physics and falsifiable error budgets.

11. Final upgraded definition

UGTS-KC 2.0 defines the substrate as a finite, queryable algebra over typed geometry, topology, kinematics and bounded dynamics. Patterns are parameterized objects with frames and derivative contracts. Topology is represented by explicit gluing maps, route descriptors and incidence invariants. Kinematics evaluates state and moving frames without requiring a frame loop for supported closed expressions. Dynamics is admitted through measured or numerically bounded flows. Events remain the sole authority for discrete change.

UGTS-KC 2.0 = (G, P, F, K, D, R, S, C, T, I, L)

G : finite grammar
 P : parametric pattern family
 F : implicit field / surface family
 K : kinematic calculus and frame operators
 D : bounded continuous or hybrid dynamics
 R : relation and guard family
 S : local support predicates
 C : compatibility predicates
 T : transition and patch family
 I : invariants / topology descriptors
 L : lineage and external novelty log

The canonical shorthand is therefore expanded, but not replaced:

CANONICAL SHORTHAND

pattern/field + kinematic state -> local support -> compatibility -> guard classification -> verified event ->
 route/topology/dynamic reset -> lineage + novelty log

The upgrade is technically coherent to the extent demonstrated by the bundled schema, implementation and tests. Performance, physical advantage and application-specific numerical certification remain future measurements.

Appendix A. M198-M257 extended mechanism catalog

These 60 rows are engineering additions. They extend, but do not rewrite, the 197-mechanism baseline. Validation labels refer only to the files shipped in this package.

ID	Domain	Mechanism	Definition and formula	Substrate integration	Validation
M198	Pattern geometry	Superellipse (Lame curve)	Parametric constant-exponent family $ x/a ^n + y/b ^n = 1$ with explicit sign restoration. Formula: $x=a*\text{sgn}(\cos t) \cos t ^{2/n}$; $y=b*\text{sgn}(\sin t) \sin t ^{2/n}$	Pattern primitive; may define support, route boundary or swept profile.	Implemented + tested
M199	Pattern geometry	Gielis superformula	Bounded radial pattern grammar that generalizes circles, polygons, stars and lobed forms. Formula: $r=(\cos(m t/4)/a ^n2 + \sin(m t/4)/b ^n3)^{(-1/n1)}$	Versioned parameter tuple; denominator floor prevents singular numerical blow-up.	Implemented + tested
M200	Pattern geometry	Rose curve	Polar harmonic petal pattern with integer or rational angular frequency. Formula: $r=a \cos(k t)$	Periodic support/template; winding and event surfaces remain separately typed.	Implemented + tested
M201	Pattern geometry	Lissajous curve	Two-axis harmonic pattern with declared frequencies and phase offset. Formula: $x=A \sin(a t+\delta)$; $y=B \sin(b t)$	Kinematic test path, phase compatibility probe or calibration trajectory.	Implemented + tested
M202	Pattern geometry	Epitrochoid	Point attached to a circle rolling outside another circle. Formula: $x=(R+r)\cos t-d \cos((R+r)t/r)$; $y=(R+r)\sin t-d \sin((R+r)t/r)$	Closed periodic route when radius ratio is rational; otherwise bounded quasi-periodic sample.	Implemented + tested
M203	Pattern geometry	Hypotrochoid	Point attached to a circle rolling inside another circle. Formula: $x=(R-r)\cos t+d \cos((R-r)t/r)$; $y=(R-r)\sin t-d \sin((R-r)t/r)$	Pattern route with cusp/self-intersection events classified by derivative and sheet.	Implemented + tested
M204	Pattern geometry	Cycloid	Trace of a point on a circle rolling along a line. Formula: $x=r(t-\sin t)$; $y=r(1-\cos t)$	Exact benchmark trajectory for cusp and contact events.	Implemented + tested
M205	Pattern geometry	General trochoid	Rolling-circle trace with point offset d distinct from radius r. Formula: $x=r t-d \sin t$; $y=r-d \cos t$	Prolate/curtate transition controlled by d/r and classified at d=r.	Implemented + tested
M206	Pattern geometry	Circle involute	Taut-string unwrapping curve with monotonic arc parameter. Formula: $x=r(\cos t+t \sin t)$; $y=r(\sin t-t \cos t)$	Gear/contact profile and nonuniform curvature benchmark.	Implemented + tested
M207	Pattern geometry	Catenary	Static hanging-chain profile retained as a bounded mechanical shape, not a generic parabola. Formula: $y=a \cosh((x-x_0)/a)+y_0$	Constraint shape or potential-energy equilibrium reference.	Implemented + tested
M208	Pattern geometry	Clothoid / Euler spiral	Curve whose curvature changes linearly with arc length. Formula: $\kappa(s)=s/A^2$; $x=\int \cos(u^2/2 A^2) d u$; $y=\int \sin(u^2/2 A^2) d u$	Curvature-continuous route connector and steering transition.	Implemented numerically + tested
M209	Pattern geometry	Fermat spiral	Square-root radial spiral with explicit branch sign. Formula: $r=a \sqrt{t}(t)$, branch selects sign/phase	Two-arm address pattern and bounded sample schedule.	Implemented + tested

ID	Domain	Mechanism	Definition and formula	Substrate integration	Validation
M210	Pattern geometry	Archimedean spiral	Linear radial growth under angular phase. Formula: $r=a+b\ t$	Uniform shell-crossing benchmark and address sweep.	Implemented + tested
M211	Pattern geometry	Rational quadratic Bezier conic	Three control points with middle weight representing exact conic arcs under valid weights. Formula: $B(t)=\frac{(1-t)^2P_0+2wt(1-t)P_1+t^2P_2}{[(1-t)^2+2wt(1-t)+t^2]}$	Conic-support primitive with denominator guard and schema-bound weight.	Implemented + tested
M212	Pattern geometry	Cubic Bezier segment	Four-point polynomial curve with endpoint tangent control. Formula: $B(t)=(1-t)^3P_0+3(1-t)^2tP_1+3(1-t)t^2P_2+t^3P_3$	Glyph morph, route segment and local sweep centerline.	Implemented + tested
M213	Pattern geometry	Open-uniform B-spline	Piecewise polynomial curve with local control and declared degree/knot vector. Formula: $C(t)=\sum_i N_{i,p}(t) P_i$	Finite grammar output normalized to bounded control points and degree.	Implemented + tested
M214	Pattern geometry	NURBS curve	Rational B-spline with explicit positive weights and knot vector. Formula: $C(t)=\frac{\sum_i N_{i,p} w_i P_i}{\sum_i N_{i,p} w_i}$	Exact/approximate CAD-style route with denominator and weight validation.	Implemented + tested
M215	Pattern geometry	Reuleaux constant-width profile	Three circular arcs producing a convex constant-width curve. Formula: Boundary assembled from radius-w arcs centered at equilateral-triangle vertices.	Constant-clearance support profile and rotation benchmark.	Implemented sampler + tested
M216	Pattern geometry	Voronoi / power-cell selector	Nearest-site partition generalized by additive site weights. Formula: $\operatorname{argmin}_i x-s_i ^2-w_i$	Compatibility/routing partition compiled to local candidate tests.	Implemented + tested
M217	Pattern geometry	Medial-axis skeletal approximation	Centerline candidates derived from equal-distance pairs or sampled boundary normals. Formula: $d(x,A)=d(x,B)$ with local maximality condition	Compressed shape skeleton; approximation status and sampling density are logged.	Implemented sampled helper + tested
M218	Field/surface geometry	Superquadric implicit field	Exponent-controlled 3D generalization of ellipsoids and box-like rounded solids. Formula: $((x/a ^{2/e_2}+ y/b ^{2/e_2})^{e_2/e_1}+ z/c ^{2/e_1})-1$	Implicit event/support field; not automatically an exact Euclidean SDF.	Implemented + tested
M219	Field/surface geometry	Anisotropic ellipsoid field	Quadratic form with positive scales or covariance-like metric. Formula: $f(x)=x^T A x-1$, A positive definite	Directional guard/support with explicit metric and conditioning checks.	Implemented + tested
M220	Field/surface geometry	Swept tube field	Distance-to-centerline field minus radius using a sampled or analytic curve. Formula: $f(x)=\min_s x-c(s) -r(s)$	Cable, vessel, wormhole-route visualization or collision support; sampled error is declared.	Implemented sampled + tested
M221	Field/surface geometry	Generalized cylinder / profile sweep	Cross-section transported along a centerline with an explicit moving frame. Formula: $X(s,u,v)=c(s)+u\ e_1(s)+v\ e_2(s)$	Pattern extrusion using Frenet or Bishop frame; frame choice is schema-bound.	Implemented helper + example

ID	Domain	Mechanism	Definition and formula	Substrate integration	Validation
M222	Field/surface geometry	Ruled surface	Surface formed by linearly connecting corresponding points on two curves. Formula: $X(u,v)=(1-v)c_0(u)+v\ c_1(u)$	Portal sheet, transition membrane or developable approximation.	Implemented + tested
M223	Field/surface geometry	Helicoid	Ruled minimal-surface family generated by screw motion. Formula: $X(u,v)=(v\ \cos u, v\ \sin u, p\ u)$	Screw-symmetry route/sheet example; physical minimality is not assumed in discrete realization.	Implemented + tested
M224	Field/surface geometry	Catenoid	Surface of revolution of a catenary. Formula: $X(u,v)=(a\ \cosh(v/a)\cos u, a\ \cosh(v/a)\sin u, v)$	Neck/pinch geometry with finite guard and no singular-energy claim.	Implemented + tested
M225	Field/surface geometry	Gyroid periodic field	Triply periodic implicit pattern used as a bounded field template. Formula: $\sin x\ \cos y + \sin y\ \cos z + \sin z\ \cos x = c$	Periodic support/route field; topology depends on threshold and finite domain.	Implemented + tested
M226	Field/surface geometry	Schwarz-P periodic field	Triply periodic cosine field serving as a connected-sheet template. Formula: $\cos x+\cos y+\cos z=c$	Field grammar primitive with threshold and cell count explicitly bounded.	Implemented + tested
M227	Field/surface geometry	Metaball / R-function blend	Smooth influence-field combination with explicit compact or inverse-distance kernel. Formula: $f(x)=\sum_i w_i\ \phi(\ x-c_i \ \)-\tau$	Merge/split event benchmark; kernel support and saturation are declared.	Implemented compact kernel + tested
M228	Field/surface geometry	Morphological offset / dilation	Parallel-set transform expressed as a field offset when the input is distance-like. Formula: $f_{\ \delta}(x)=f(x)-\delta$	Guard expansion, tolerance shell and manufacturing clearance; exact only for valid distance fields.	Implemented + tested
M229	Field/surface geometry	Wavefront / eikonal arrival field	Arrival-time field under a declared local propagation speed. Formula: $ \ \text{grad}\ T =1/c(x)$; constant-c case $T=\text{distance}/c$	Event-time support primitive; variable-speed solver is specified separately.	Implemented constant-speed + tested
M230	Topology	Winding number / degree tag	Integer-valued enclosure/orientation descriptor for closed planar routes. Formula: $w=(1/2\pi)\ \text{integral}\ d\ \arg(\gamma(t)-x)$	Inside/outside and route class metadata without collapsing geometry to a bit.	Implemented polygonal + tested
M231	Topology	Covering-space lift	Unwrapped coordinate retains traversal count across periodic quotient boundaries. Formula: $\text{theta_lift}=\text{theta}+2\pi\ k$	Prevents ambiguity at repeated coordinates and supports lineage reconstruction.	Implemented + tested
M232	Topology	Homotopy-class route label	Route label stores obstacle winding/signature rather than only endpoint coordinates. Formula: $h=(w_1,...,w_n)$ for declared obstacle generators	Compatibility and branch selection may require matching or allowed classes.	Implemented descriptor + tested
M233	Topology	Braid-word transition	Ordered crossings represented by generators σ_i and inverses with local cancellation. Formula: $\dots\ \sigma_i\ \sigma_i^{-1}\ \dots\ \rightarrow$ identity	Multi-entity route lineage; word length is bounded and reduction policy explicit.	Implemented reducer + tested

ID	Domain	Mechanism	Definition and formula	Substrate integration	Validation
M234	Topology	Knot/link checksum	Optional low-cost invariant or numerical linking estimate for closed route pairs. Formula: Gauss linking integral or declared discrete approximation	Diagnostic checksum only; not complete knot identity and not used as sole authority.	Specified; numerical certification pending
M235	Topology	Cell-complex incidence	Vertices, edges, faces and cells connected through oriented incidence maps. Formula: boundary_k: C_k -> C_{k-1}	Topology storage independent of embedding; supports local gluing and route updates.	Implemented boundary matrices + tested
M236	Topology	Boundary/coboundary update	Topology transition updates primal incidence and dual cochain state consistently. Formula: boundary_{k-1} boundary_k = 0	Validation rejects edits violating boundary-of-boundary zero.	Implemented invariant check + tested
M237	Topology	Discrete Morse critical event	Cell pairing identifies local critical cells where topology/routing may change. Formula: Unpaired cells are critical under a declared matching.	Event candidate for simplification or handle birth/death; matching is explicit.	Implemented descriptor + example
M238	Topology	Persistent-homology summary	Birth/death intervals summarize bounded filtration changes. Formula: barcode={birth_i,death_i,dimension_i}	Diagnostic feature vector; does not replace full geometry or lineage.	Specified exchange format; solver not bundled
M239	Topology	Atlas transition cocycle	Overlapping chart maps are checked for consistent composition. Formula: g_{ij} g_{jk} = g_{ik} on triple overlaps	Prevents incompatible portal/gluing definitions in multi-chart worlds.	Implemented matrix check + tested
M240	Kinematic calculus	SE(2) rigid transform and twist	Planar position/orientation update represented by the exponential of a body twist. Formula: T(t)=exp(t [v_x,v_y,omega]^)	Exact constant-twist state_at(t) and portal patch primitive.	Implemented + tested
M241	Kinematic calculus	SE(3) twist exponential	Spatial rigid motion generated by angular and linear velocity. Formula: T=exp([omega,v]^ t) with Rodrigues rotation and V matrix	Constant screw motion, frame transport and analytic state query.	Implemented + tested
M242	Kinematic calculus	Quaternion SLERP	Shortest-arc interpolation of unit orientations with antipodal handling. Formula: q(t)=sin((1-t)theta)/sin theta q0 + sin(t theta)/sin theta q1	Orientation interpolation separate from position and topology.	Implemented + tested
M243	Kinematic calculus	Screw interpolation (SE(3) geodesic)	Interpolate through the relative rigid transform log/exponential rather than independent Euler angles. Formula: T(t)=T0 exp(t log(T0^-1 T1))	Portal/coupler motion with consistent rotation-translation coupling.	Implemented + tested
M244	Kinematic calculus	Frenet-Serret moving frame	Tangent, normal and binormal derived from nondegenerate curve derivatives. Formula: T=r'/ r' ; B=(r' x r'')/ r' x r'' ; N=B x T	Curvature/torsion diagnostics and sweep orientation where curvature is nonzero.	Implemented + tested

ID	Domain	Mechanism	Definition and formula	Substrate integration	Validation
M245	Kinematic calculus	Bishop parallel-transport frame	Rotation-minimizing frame transported along a sampled curve. Formula: Project previous normal into new tangent plane, then renormalize.	Stable sweep frame through near-zero-curvature segments.	Implemented + tested
M246	Kinematic calculus	Arc-length reparameterization	Monotone lookup maps path parameter to cumulative distance and back. Formula: $s(t)=\int_0^t r'(u) du$	Uniform-speed queries and event bracketing with declared interpolation error.	Implemented sampled + tested
M247	Kinematic calculus	Curvature-limited steering	Speed and angular rate constrained by path curvature and lateral acceleration. Formula: $\omega=v \kappa$; $v \leq \sqrt{(a_{lat_max}/\max(\kappa , \epsilon))}$	Feasible trajectory gate before event solving.	Implemented + tested
M248	Kinematic calculus	Jerk-bounded quintic S-curve	Endpoint-rest time law with continuous position, velocity and acceleration. Formula: $s(u)=10u^3-15u^4+6u^5$, u in $[0,1]$	Smooth phase/route actuation; peak derivatives determine duration contract.	Implemented + tested
M249	Kinematic calculus	Limit-aware time scaling	Choose trajectory duration so normalized velocity, acceleration and jerk stay within declared limits. Formula: $T=\max(D*v_{peak}/v_{max}, \sqrt{D*a_{peak}/a_{max}}, \sqrt[3]{D*j_{peak}/j_{max}})$	Bounded schedule; not claimed globally time-optimal for arbitrary paths.	Implemented + tested
M250	Kinematic calculus	Articulated-chain forward kinematics	Compose joint transforms to obtain link and endpoint frames. Formula: $T_{end}=\text{product}_i T_i(q_i)$	Finite grammar may emit link chains; state query returns all frames and lineage.	Implemented planar chain + tested
M251	Kinematic calculus	Jacobian damped least-squares IK	Regularized local inverse step avoids singular matrix blow-up. Formula: $\Delta q=J^T(JJ^T+\lambda I)^{-1} e$	Optional solver with residual, iteration budget and failure status.	Implemented planar chain + tested
M252	Dynamics and events	Symplectic / semi-implicit integration	Velocity is updated before position for improved qualitative energy behavior in separable mechanics. Formula: $v_{(n+1)}=v_n+a(x_n)dt$; $x_{(n+1)}=x_n+v_{(n+1)}dt$	Fallback dynamic state update; timestep and error budget remain explicit.	Implemented + tested
M253	Dynamics and events	Damped second-order mode	Mass-spring-damper evolution with typed mass, stiffness and damping. Formula: $m x''+c x'+k x=f(t)$	Actuator, hinge or threshold reset dynamics; energy bookkeeping is measurable.	Implemented + tested
M254	Dynamics and events	Graph-Laplacian diffusion / coupled modes	Local state exchange over an explicit adjacency graph. Formula: $x'=-\alpha L x$ or $x''+C x'+Kx=0$	Phase smoothing, consensus or coupled oscillation without hidden global field.	Implemented first-order diffusion + tested
M255	Dynamics and events	Reaction-diffusion pattern dynamics	Two-field local reaction plus diffusion update on a bounded grid or graph. Formula: Gray-Scott: $U_t=D_u \text{ Lap } U-UV^2+F(1-U)$; $V_t=D_v \text{ Lap } V+UV^2-(F+k)V$	Dynamic pattern generator kept downstream of typed field/state schema.	Implemented grid step + tested

ID	Domain	Mechanism	Definition and formula	Substrate integration	Validation
M256	Dynamics and events	Hybrid automaton continuous/discrete dynamics	Each mode has a continuous flow; guard crossings invoke discrete reset and route changes. Formula: $x'=f_m(x)$; if $g_{(m,n)}(x)=0$ then $x^+=R_{(m,n)}(x)$	Generalizes threshold cup, parity hinge and event transition into a typed model.	Implemented reference class + tested
M257	Dynamics and events	Event degeneracy and simultaneity resolver	Classify crossing, tangency and coincidence; resolve equal-time patches by priority or verified commutativity. Formula: Use sign history, derivative, residual interval and patch-conflict graph.	Prevents chatter, unordered lineage and silent overwrite at multi-guard events.	Implemented classifier/resolver + tested

Appendix B. API and exchange dictionary

B.1 Pattern and field functions

Function	Returns	Contract
superellipse(theta,a,b,n)	2D point	Positive a,b,n.
gielis(theta,m,n1,n2,n3,a,b)	2D point	Denominator floor; bounded parameters.
clothoid(s,A,steps)	2D point	Numerical quadrature; steps in validation contract.
bspline / nurbs(control,...)	N-D point	Degree, knots and weights validated.
superquadric_field(p,scales,e1,e2)	scalar	Implicit field, not exact SDF by default.
gyroid_field / schwarz_p_field	scalar	Threshold and finite domain required.
tube_field(p,samples,radius)	scalar	Sampled centerline error declared.

B.2 Topology and kinematics

Function	Returns	Contract
winding_number(route,point)	real near integer	Closed polygonal route; sign carries orientation.
homotopy_signature(route,obstacles)	integer tuple	Obstacle generators are declared by schema.
CellComplex.boundary_squared_zero()	bool	Checks incidence invariant only.
se2_exp / se3_exp	homogeneous transform	Constant body twist.
se3_interpolate(T0,T1,t)	homogeneous transform	Relative log/exponential interpolation.
frenet_frame(r1,r2)	T,N,B,kappa	Requires nonzero velocity and curvature.
bishop_frames(points,normal)	frame list	Sampled rotation-minimizing transport.
quintic_scurve(u,D,T)	position,velocity,accel,jerk	u clamped to [0,1].
dls_ik(lengths,q0,target)	angles,end point	Bounded iterations and residual.

B.3 Dynamics and event resolution

Function	Returns	Contract
symplectic_euler	state step	dt and acceleration callback explicit.
graph_diffusion_step	state tuple	Edges and weights explicit; stability is caller contract.
gray_scott_step	two grids	Periodic boundary in reference implementation.
HybridAutomaton.step	state + mode	Transition priority and reset functions explicit.
bisect_root	root estimate	Requires sign bracket.

Function	Returns	Contract
classify_event	status string	Uses before/at/after values and optional derivative.
resolve_patches	merged patch + conflicts	Priority first; differing values produce conflict record.

B.4 Exchange schema notes

- Schema version is 2.0.0.
- Attribution fields are included with verification_status=requester-supplied-not-independently-verified.
- Pattern instances reference mechanism IDs M198-M257 and carry parameter and validation objects.
- Dynamics instances declare closed_form, twist, s_curve, integrated or hybrid mode.
- Projection settings remain non-authoritative downstream metadata.

Appendix C. Source and package verification

C.1 Source basis

Document	Pages	SHA-256	Included in ZIP
UGTS-0 1.0	45	c991aba24cc939ba680697b62b9e1 9545131cf34060793490bf149c6e2a b522e	false
UGTS-GN 1.1	28	3a6b6c57118ca4e2d7d7e577508c3 477a095d133791f3708cb56a05428 5225d2	false

The source PDFs are not redistributed in the package. Their hashes allow the requester to verify which supplied versions were used.

C.2 Package verification

- M198-M257 were counted at 60 and the extended total at 257.
- 47 unit tests passed; captured output is included.
- All diagrams in the report were generated from the bundled functions or schematic drawing code.
- A SHA-256 manifest covers the delivered package files.
- The PDF was rendered from the editable DOCX and visually inspected before ZIP creation.

C.3 Closing statement

The original corpus becomes stronger when patterns are admitted through types and bounds rather than metaphor. Geometry defines curves, fields and supports. Topology defines explicit route classes, gluing and incidence. Kinematics defines frames and state-at-time. Dynamics defines measurable flows and resets. The event calculus remains the authority that turns those structures into verified state change.